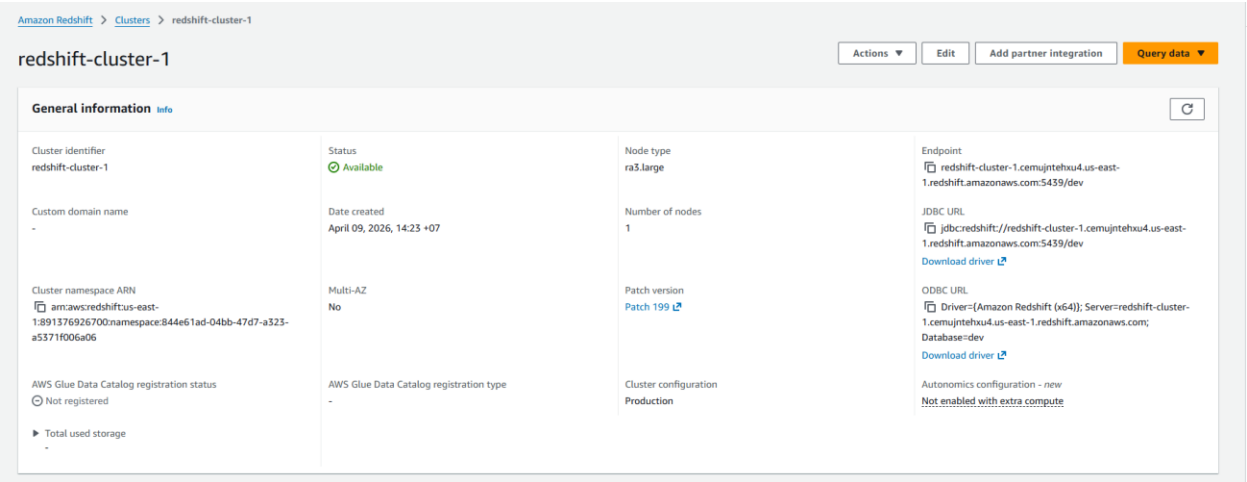
Creation of a Redshift Cluster

I. Screenshots of the configuration of the Redshift cluster that you have created:



Setting up a database in the Redshift cluster and running queries to create the dimension and fact tables

II. Queries to create the various dimension and fact tables with appropriate primary and foreign keys:

Note: The dimension tables were created with appropriate primary keys, and the fact table was designed with foreign key columns that logically reference the corresponding dimension tables. Although Amazon Redshift does not enforce referential integrity constraints, the relationships are maintained at the data modeling level.

1. dim_location

```
CREATE TABLE dim_location (
   location_id INT PRIMARY KEY,
   location VARCHAR(50),
   streetname VARCHAR(255),
   street_number INT,
   zipcode INT,
   lat DECIMAL(10,3),
   lon DECIMAL(10,3)
);
```

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Summary

Returned rows: 0

Elapsed time: 424ms

Result set query:

```
/* RQEV2-YPoungF1pL */
CREATE TABLE dim_location (
    location_id INT PRIMARY KEY,
    location VARCHAR(50),
    streetname VARCHAR(255),
    street_number INT,
    zipcode INT,
    lat DECIMAL(10,3),
    lon DECIMAL(10,3)
)
```

2. **dim_atm**

```
CREATE TABLE dim_atm (
    atm_id INT PRIMARY KEY,
    atm_number VARCHAR(20),
    atm_manufacturer VARCHAR(50),
    atm_location_id INT
);
```

Summary

Returned rows: 0

Elapsed time: 474ms

Result set query:

```
/* RQEV2-ANulZDAD9G */
CREATE TABLE dim_atm (
    atm_id INT PRIMARY KEY,
    atm_number VARCHAR(20),
    atm_manufacturer VARCHAR(50),
    atm_location_id INT
)
```

3. **dim_date**

```
CREATE TABLE dim_date (
    date_id INT PRIMARY KEY,
    full_date_time TIMESTAMP,
    year INT,
    month VARCHAR(20),
    day INT,
    hour INT,
    weekday VARCHAR(20)
```

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);

Summary

Returned rows: 0

Elapsed time: 409ms

Result set query:

```
/* RQEV2-0JkgFmkpA2 */
CREATE TABLE dim_date (
    date_id INT PRIMARY KEY,
    full_date_time TIMESTAMP,
    year INT,
    month VARCHAR(20),
    day INT,
    hour INT,
    weekday VARCHAR(20)
)
```

4. **dim_card_type**

```
CREATE TABLE dim_card_type (
    card_type_id INT PRIMARY KEY,
    card_type VARCHAR(30)
);
```

Summary

Returned rows: 0

Elapsed time: 349ms

Result set query:

```
/* RQEV2-8bGua2b2N7 */
CREATE TABLE dim_card_type (
    card_type_id INT PRIMARY KEY,
    card_type VARCHAR(30)
)
```

5. **fact_atm_trans**

```
CREATE TABLE fact_atm_trans (
    trans_id BIGINT PRIMARY KEY,
    atm_id INT,
    weather_loc_id INT,
    date_id INT,
    card_type_id INT,
    atm_status VARCHAR(20),
    currency VARCHAR(10),
    service VARCHAR(20),
    transaction_amount INT,
    message_code VARCHAR(255),
    message_text VARCHAR(255),
```

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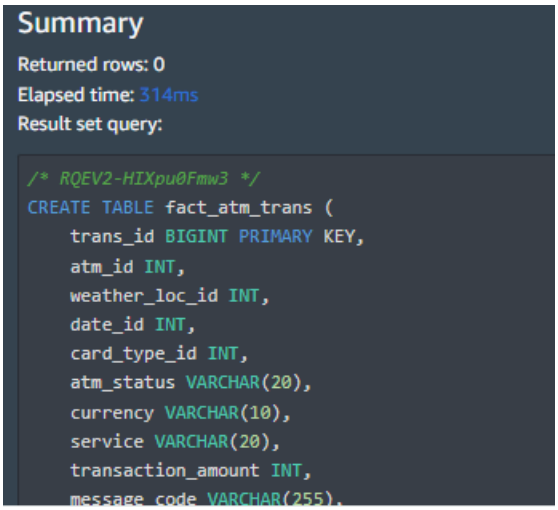
```
    rain_3h DECIMAL(10,3),
    clouds_all INT,
    weather_id INT,
    weather_main VARCHAR(50),
    weather_description VARCHAR(255)
);
```



Loading data into a Redshift cluster from Amazon S3 bucket

III. Queries to copy the data from S3 buckets to the Redshift cluster in the appropriate tables

Note: Due to strict Parquet schema enforcement in Redshift COPY operations, Redshift Spectrum was used as a staging mechanism to load the data from S3 into Redshift tables. This approach still fulfills the requirement of copying data from S3 to Redshift.

```
CREATE EXTERNAL SCHEMA ext_schema
FROM DATA CATALOG
DATABASE 'default'
IAM_ROLE 'arn:aws:iam::891376926700:role/service-role/AmazonRedshift-
CommandsAccessRole-20260409T141751'
CREATE EXTERNAL DATABASE IF NOT EXISTS;
```

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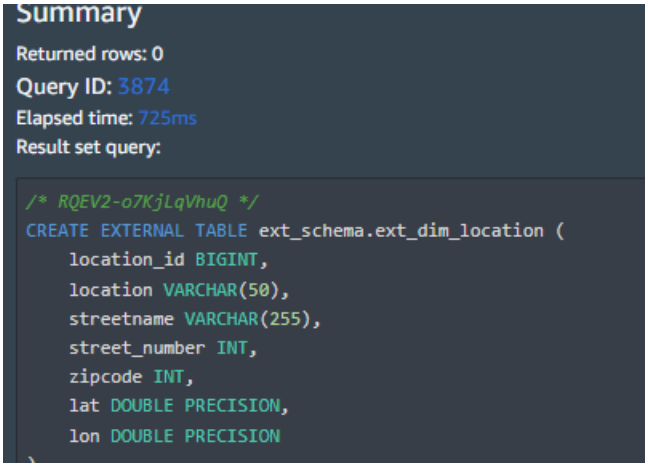
1. **dim_location**

```
CREATE EXTERNAL TABLE ext_schema.ext_dim_location (
    location_id BIGINT,
    location VARCHAR(50),
    streetname VARCHAR(255),
    street_number INT,
    zipcode INT,
    lat DOUBLE PRECISION,
    lon DOUBLE PRECISION
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_location/';
```




```
INSERT INTO dim_location
SELECT
    location_id,
    location,
    streetname,
    street_number,
    zipcode,
    CAST(lat AS DECIMAL(10,3)),
    CAST(lon AS DECIMAL(10,3))
FROM ext_schema.ext_dim_location;
```

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Summary

Affected rows: 109

Query ID: 3932

Elapsed time: 1.2s

Result set query:

```
/* RQEV2-AeoSlSSWQL */
INSERT INTO dim_location
SELECT
    location_id,
    location,
    streetname,
    street_number,
    zipcode,
    CAST(lat AS DECIMAL(10,3)),
    CAST(lon AS DECIMAL(10,3))
FROM ext_schema.ext_dim_location
```

2. **dim_atm**

```
CREATE EXTERNAL TABLE ext_schema.ext_dim_atm (
    atm_id INT,
    atm_number VARCHAR(20),
    atm_manufacturer VARCHAR(50),
    atm_location_id BIGINT
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_atm/';
```

Summary

Returned rows: 0

Query ID: 3932

Elapsed time: 597ms

Result set query:

```
/* RQEV2-EqClT5U6LZ */
CREATE EXTERNAL TABLE ext_schema.ext_dim_atm (
    atm_id INT,
    atm_number VARCHAR(20),
    atm_manufacturer VARCHAR(50),
    atm_location_id BIGINT
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_atm/'
```

```
INSERT INTO dim_atm
SELECT
    atm_id,
    atm_number,
```

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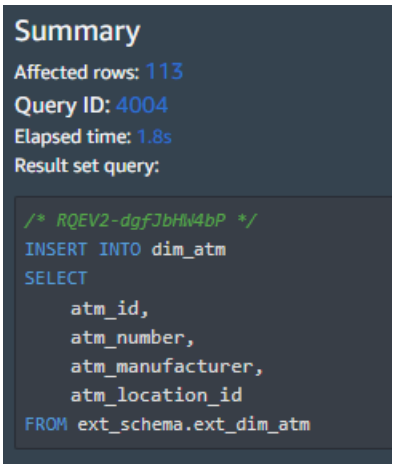
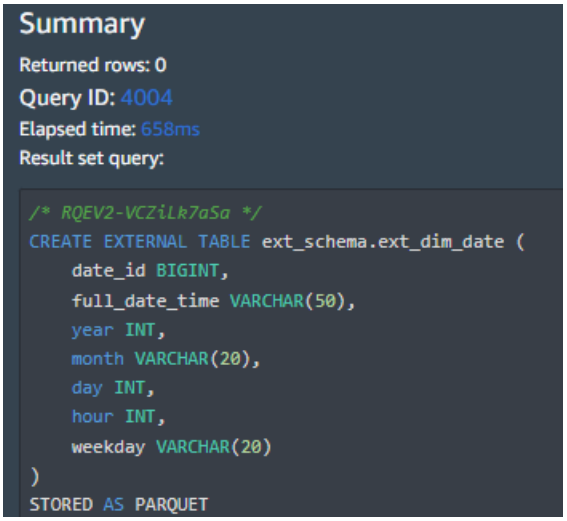
```
    atm_manufacturer,
    atm_location_id
FROM ext_schema.ext_dim_atm;
```



3. **dim_date**

```
CREATE EXTERNAL TABLE ext_schema.ext_dim_date (
    date_id BIGINT,
    full_date_time VARCHAR(50),
    year INT,
    month VARCHAR(20),
    day INT,
    hour INT,
    weekday VARCHAR(20)
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_date/';
```



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```
INSERT INTO dim_date
SELECT
    date_id,
    TO_TIMESTAMP(full_date_time, 'YYYY Month DD HH24'),
    year,
    month,
    day,
    hour,
    weekday
FROM ext_schema.ext_dim_date;
```

Summary

Affected rows: 8685
Query ID: 4083
Elapsed time: 2s
Result set query:

```
/* RQEV2-ueML2sAALo */
INSERT INTO dim_date
SELECT
    date_id,
    TO_TIMESTAMP(full_date_time, 'YYYY Month DD HH24'),
    year,
    month,
    day,
    hour,
    weekday
```

4. **dim_card_type**

```
CREATE EXTERNAL TABLE ext_schema.ext_dim_card_type (
    card_type_id BIGINT,
    card_type VARCHAR(30)
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_card_type/';
```

Summary

Returned rows: 0
Query ID: 4083
Elapsed time: 553ms
Result set query:

```
/* RQEV2-gKjPj6UOyT */
CREATE EXTERNAL TABLE ext_schema.ext_dim_card_type (
    card_type_id BIGINT,
    card_type VARCHAR(30)
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/dim_card_type/'
```

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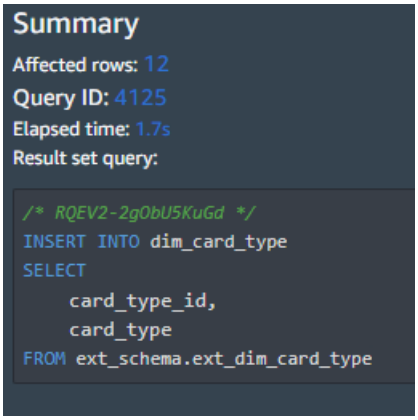
```
INSERT INTO dim_card_type
SELECT
    card_type_id,
    card_type
FROM ext_schema.ext_dim_card_type;
```



5. **fact_atm_trans**

```
CREATE EXTERNAL TABLE ext_schema.ext_fact_atm_trans (
    trans_id BIGINT,
    atm_id INT,
    weather_loc_id BIGINT,
    date_id BIGINT,
    card_type_id BIGINT,
    atm_status VARCHAR(20),
    currency VARCHAR(10),
    service VARCHAR(20),
    transaction_amount INT,
    message_code VARCHAR(255),
    message_text VARCHAR(255),
    rain_3h DOUBLE PRECISION,
    clouds_all INT,
    weather_id INT,
    weather_main VARCHAR(50),
    weather_description VARCHAR(255)
)
STORED AS PARQUET
LOCATION 's3://mike-upgrad-etl-project/fact_atm_trans/';
```

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Summary

Returned rows: 0

Query ID: 4125

Elapsed time: 491ms

Result set query:

```
/* RQEV2-0nb3Fh35vm */
CREATE EXTERNAL TABLE ext_schema.ext_fact_atm_trans (
    trans_id BIGINT,
    atm_id INT,
    weather_loc_id BIGINT,
    date_id BIGINT,
    card_type_id BIGINT,
    atm_status VARCHAR(20),
    currency VARCHAR(10),
    service VARCHAR(20),
```

```
INSERT INTO fact_atm_trans
SELECT
    trans_id,
    atm_id,
    weather_loc_id,
    date_id,
    card_type_id,
    atm_status,
    currency,
    service,
    transaction_amount,
    message_code,
    message_text,
    CAST(rain_3h AS DECIMAL(10,3)),
    clouds_all,
    weather_id,
    weather_main,
    weather_description
FROM ext_schema.ext_fact_atm_trans;
```

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Summary

Affected rows: 2468572

Query ID: 4151

Elapsed time: 6.6s

Result set query:

```
/* RQEV2-mEzDg4cmaC */
INSERT INTO fact_atm_trans
SELECT
    trans_id,
    atm_id,
    weather_loc_id,
    date_id,
    card_type_id,
    atm_status,
    currency,
    service
```

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